shreyas ml.docx

Birla Institute of Technology, Mesra

Document Details

Submission ID

trn:oid:::3117:536372249

Submission Date

Dec 4, 2025, 9:41 PM GMT+5:30

Download Date

Dec 4, 2025, 9:43 PM GMT+5:30

File Name

shreyas ml.docx

File Size

32.9 KB

8 Pages

4,759 Words

27,383 Characters

*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Disclaimer
Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?
The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?
Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

Intelligent Disease Surveillance: Machine Learning-Driven Early Detection of Vector-Borne Disease Outbreaks Using Climate and Health Data

Shivani A
Dept of AI-ML
Dayananda Sagar University
Bengaluru, Karnataka, India
eng23am0070@dsu.edu.in

Shreyas D C
Dept of AI-ML
Dayananda Sagar University
Bengaluru, Karnataka, India
eng23am0071@dsu.edu.in

Trupthi Rao
Dept of AI-ML
Dayananda Sagar University
Bengaluru, Karnataka, India
trupthirao-aiml@dsu.edu.in

Sneha B
Dept of AI-ML
Dayananda Sagar University
Bengaluru, Karnataka, India
eng23am0073@dsu.edu.in

Spoorthi
Dept of AI-ML
Dayananda Sagar University
Bengaluru, Karnataka, India
eng23am0074@dsu.edu.in

INTRODUCTION

Vector-borne illnesses such as malaria, dengue, or Zika should not be a major source of public health concern. These diseases are typical of regions with a hot and humid climate all year round or a place that has both wet and dry seasons. Such diseases are caused by vectors, e.g., mosquitoes, which in most cases, after every change of weather, come up with new habitats. Health-related information, including the number of reported cases and demographic data, facilitates understanding disease transmission and helps in the identification of the risk areas. The use of both health and climate data provides the opportunity for researchers t Vector-borne diseases like malaria, dengue, or Zika virus should not be a major public health issue generally. These diseases are characteristic of areas having either a hot and humid climate all year or a place that has both wet and dry seasons. Such illnesses are the result of vectors e.g., mosquitoes, which in most cases after every change in weather, reap the benefits of the new habitats. Demographic data on the number of reported cases in conjunction with health-related information support the understanding of the disease transmission and help in identifying the risk areas. Health and climate data together offer an opening for scientists to construct models that indicate the future and thus give the time and place of the epidemics, their monitoring, and the preparedness for the response. This composite approach is the center of the platforms for the earliest communication of the emergency and the most efficient local public health steps for the repositioning of human resources.

The death of the worldwide number of people due to malaria was approximately half a million (around 608,000) in 2022, while the total number of malaria cases was close to 249 million – these statistics have largely dominated the estimated figures before the COVID-19 pandemic, and there has been an increase of five million cases since 2021.12. Since October 2021, WHO has been promoting mainly children the use of the RTS,S/AS01 malaria vaccine in areas with moderate to high P. falciparum malaria prevalence.

The global incidence of dengue has been on a steady climb over the last several years. The number of cases reported to the WHO has been gradually rising from 505,430 in 2000 to 6.5 million in 2023. Almost half of the world's population is at the risk. Nevertheless, health information such as disease occurrence, demographic changes, and healthcare access, etc. are the factors that reveal population vulnerability and gradual disease transmission. By utilizing both climate and health data, scientists and public health departments can follow the trends, develop predictive models, and receive prompt notifications of the outbreak. The research-based solution to this problem is

the pioneer of early warning systems, thus giving the intervention the appropriate kind and time, the resources being in the place of need, and public health strategic planning being well thought out – hence the decrease of the vector-borne diseases' threat and the saving of lives.

II.LITERATURE REVIEW

An investigation conducted by Andrew K. Githeko,[1] besides the factors that have already been confirmed leading to vector-borne diseases i.e. seasonal variation of weather, socioeconomic status... moreo reate models that help predict the future and thus give a forecast of the outbreaks, their monitoring, and the readiness to respond. This integrated method is at the core of the platforms for the earliest communication of the emergency and the most effective local public health measures for human resources' repositioning.

The number of the world death toll due to malaria was about a half a million (around 608,000) in 2022, whereas the total number of malaria cases was almost 249 million – these numbers have significantly overshadowed the estimated figures before the COVID-19 pandemic, and there has been a five million increase in cases since 2021.12. Since October 2021, the WHO has been advocating the use of the RTS,S/AS01 malaria vaccine, mainly in children, in areas with moderate to high P. falciparum malaria prevalence.

The worldwide incidence of dengue has been progressively rising over the last few years. The number of cases reported to the WHO has gradually increased from 505,430 in 2000 to 6.5 million in 2023. Nearly half of the world's population is exposed to the risk. On the other hand, health information like disease occurrence, demographic changes, and healthcare access, among other things, are some of the clues of population vulnerability and gradual increments of disease transmission. By making use of both climate and health data, scientists and public health sectors can track the patterns, form predictive models, and get timely notifications of the outbreak. The research-based solution to this problem is the pioneer of early warning systems, thus giving the intervention the appropriate kind and time, the resources being in the place of need, and public health strategic planning being well thought out – hence the decrease of the vector-borne diseases' threat and the saving of lives.

II.LITERATURE REVIEW

In a study by Andrew K. Githeko,[1] in addition to the already known factors that lead to vector-borne diseases, i.e. seasonal weather variation, socioeconomic status, environmental changes and drug resistance, climate change and variability Climate change is cited by a majority of authors as the prime cause of the rise/emergence of MBDs and a global public health issue. The authors have marked the close relationships MBDs that are dependent on climate factors such as temperature, rainfall, humidity, and wind, etc. Unfortunately, due to the disparity of the studies, they failed to proceed with their planned meta-analytical research.

Holly Jarman et al. described how public health policy reform in the USA was mainly influenced by the development of legislation for mosquito control. The example of Florida demonstrated that, besides being a good visual representation of public health capacity, mosquito control also delivered real economic and personal benefits through disease limitation.

The return or recurrence of insect-borne diseases that have been traditionally affecting Latin America or were recently revealed and have been gradually spreading, is very frequently linked to human migrations and antropogenic changes such as urbanization, deforestation, and ecotourism, according to their paper. This paper presents a methodological framework that not only aids in understanding the epidemic dynamics over time but also in terms of space. The methodology provides nearly unlimited opportunities for understanding the transmission of diseases by arbovirus infections, thus it becomes feasible for public health authorities to plan and implement timely interventions for achieving better disease control.

Jonas E. Mertens[8]. Vector-borne diseases are subjected to changes in nature and climate. The results are already very diverse and quite unexpected, and they will keep on changing profoundly the distribution, transmission, and number of people at risk. Among these diseases, malaria is the most harmful one for human health - thus, the relationship between climate change and this disease is of great significance.

According to Muhammad Ammar[9], the epidemiology of diseases transmitted by mosquitoes and ticks in Pakistan is a complex and dynamic public health problem, which needs immediate attention and the implementation of control measures. The rise of diseases such as dengue, chikungunya, and Crimean–Congo haemorrhagic fever is being assisted by factors like urbanization, inadequate sanitation, climate change, and increased human–animal interactions. These factors not only facilitate the spread of already existing arboviral infections but also increase the likelihood of new pathogen outbreaks in areas that have not been infected before.

Luigi Cofone, Marise Sabato, and Carolina Di Paolo[10], The current evidence leading to future research directions should

be interdisciplinary, global, and action-oriented. Long-term studies are required to understand the effects of integrated and context-specific interventions.

F Yavari Nejad[11]. We identified a new significant risk factor that we named TRF. It was a combination of the delayed average minimum temperature over five weeks along with the current week and the delayed cumulative rainfall over two weeks. TRF has been instrumental in the prediction of dengue outbreaks and these lagged weather variables can be very useful in locating the dengue outbreak more precisely.

X Leung[12]. The major flaws in most of the present modeling attempts that have been acknowledged are the failure to utilize real-time primary predictor data, the failure to consider non-climatic parameters as predictors, and the insufficient disclosure of model development methods, model validation, and performance measure..

S Kumar[13]. The research took into consideration climate change issues and its impact on the incidences of kala-azar or VBD (vector-borne diseases). KA has been attributed as the second most significant parasitic-related cause of death worldwide with a mortality rate of 75–95%. The published literature has seriously acknowledged the rapid escalation of KA cases in 88 different endemic countries. More specifically, the origin of KA .

The paper by S Patil[14] is a prediction model for dengue incidence in various cities of Maharashtra. The model uses weather data as input and gives the predicted number of cases as output. Several meteorological parameters such as MMIN, MMAX, RH, TMRF, etc. are given as input, and the proposed system outputs the number of dengue cases.

Zheng[15]. This research delves into the promise of machine learning, especially XGBoost with SHAP values, in revealing the complex interaction of climate variables and the spread of malaria in southeastern Tanzania. The significant findings reveal that the valleys of malaria incidence are associated with seasonal changes, and among the primary reasons for district-to-district variations are temperature along with rainfall in particular.

L Hussain-Rainfall has been the primary factor behind the variability of different districts. Structural, statistical, and operational-design-related issues of EWS have also been there, which have varied with diseases and their corresponding settings, where these disparities have been reported. M. I. Leon[17]. This research combines daily dengue incidence reports from the DGHS and daily weather data from the Dark Sky API to create a dengue dataset. It was found that the dataset used in the previous studies had some missing and incorrect values, and therefore, the output produced was not very good.

AH Chowdhury[18]. The present research, more so in the climate change scenario, indicates that temperature is the leading factor as a climate factor contributing to waterborne infections in Bangladesh. The revelations demand the formation of public health plans at the district level, which are customized for the area

III METHODOLOGY

The present research employs a quantitatively and critically-analytical framework to study the effects of climate changes on the incidence of vector-borne diseases. It utilizes a retrospective method to examine secondary data that have been obtained from different reliable sources to demonstrate diseases' temporal trends, environmental changes, and spatial distribution. The application of epidemiological analysis, climate assessment, environmental evaluation, and predictive modeling is the main tool used to unravel the complex relationship between climate changes and the rise as well as the return of vector-borne diseases.

The work is concentrated in INDIA which, according to the research, is the area that has suffered the most from climate changes, environmental changes, urbanization, and has a history of the spread of vector-borne diseases. The period 2011–2024 is quite adequate for the study of both long- and short-term climatic trends. The duration is sufficient to account for variations from one year to another, extreme weather events, and significant changes in disease incidences.

Information regarding disease surveillance was obtained from national public health departments, surveillance systems, and existing databases in terms of the number of confirmed cases of the selected vector-borne diseases such as dengue, malaria, and chikungunya. These data were recorded on a weekly or monthly basis depending on the availability and were merged to correspond to the spatial and temporal resolution required for the analysis. Meteorological parameters were taken from national weather bureaus and various globally accepted databases and consisted of the average, the maximum, and the minimum temperature, the total rainfall, the relative humidity, and the wind speed. In order to signify the impact of the weather on mosquito breeding and pathogen incubation that is returned, they established time-delayed variables from one to three months for each weather parameter.

Besides weather-related variables, the researchers included environmental and socioeconomic factors in their study to have more explanatory power in their models. These were land

use and land cover patterns, the vegetation indices derived from the satellite images, closeness to water bodies, the degree of urbanization, population density, and the general housing conditions. In the event that human mobility and migration indicators are present, they were also taken into account as the movement of people can facilitate the spread of vector-borne diseases in areas that have been previously free of them.

Before the analysis was done, the researchers had already accomplished data cleaning and standardization of all the data they had collected to both verify data correctness and maintain data consistency. They eliminated duplicate entries and inconsistencies whereas for missing values they used interpolation and suitable statistical methods. They changed all the variables to a common scale so that they could be compared, and also synchronized in time the disease and climate data so that each disease observation would correspond to the respective climatic and environmental conditions. The resulting database was therefore suitable for both descriptive and predictive analysis.

To uncover the underlying trends, seasonal behaviors, and correlations between climate variables and disease incidence, the researchers carried out exploratory data analysis. They drew up timeseries charts to point out the changes in disease patterns over time and also did correlation analysis to determine the extent and the direction of the relationships between the climatic factors and the number of cases reported. Besides, they did the initial spatial visualization of the data to find the geographical clustering and areas with high incidence.

To quantify and predict disease trends, the researchers utilized conventional statistical models as well as advanced machine learning algorithms. To identify the typical associations and to forecast disease incidence based on historical trends, they used multiple linear regression and seasonal autoregressive integrated moving average (SARIMA) models. Furthermore, they implemented machine learning algorithms like Random Forest, Support Vector Machine, Gradient Boosting, and Artificial Neural Networks to detect complex non-linear interactions among climatic, environmental, and epidemiological variables. They split their data into a training set which had seventy percent of the data and a testing set which had the remaining thirty percent. During the training stage, they also used cross-validation techniques to reduce the overfitting problem and to enhance the models' generalizability.

The performance of each model was measured by using standard statistical metrics such as Root Mean Square Error, Mean Absolute Error, and the coefficient of determination. The Area Under the Receiver Operating Characteristic curve was also determined for classification-based predictions. The final outbreak prediction model was the one that demonstrated the highest predictive accuracy and stability.

Geographic Information System software was utilized for spatial analysis to pinpoint the locations that were most diseased and the clusters. The experts resorted to kernel density mapping, spatial autocorrelation, and hotspot analysis to illustrate the geographic distribution of the risk. They mixed up these spatial findings with the climatic and environmental layers to create risk maps that show the places that are more vulnerable to new outbreaks. These maps are designed to facilitate public health organization and intervention planning that is targeted at specific areas.

The research only makes use of secondary data which is publicly available and anonymized. No personal identifiers were accessed or stored during the research. The use of secondary data was in accordance with ethical standards and all the necessary permissions were granted by the respective institutions.Even though the study has carried out extensive data processing and advanced analytical techniques, it still recognizes several limitations. These limitations include possible underreporting of disease cases, differences in data quality from one region to another, and limited availability of high-resolution socioeconomic data. Nevertheless, the overall trustworthiness and scientific merit of the results are improved by the integration of various data sources and the application of robust modeling techniques despite these limitations.

Different methods of prediction varied in terms of the levels of success they achieved. Traditional regression models were able to explain overall trends and seasonal changes, but they lacked the capacity to handle complex non-linear relationships. In contrast, machine learning algorithms were found to have better predictive powers, and they even achieved the highest accuracy.

IV RESULTS

The temporal study of the occurrence of diseases led to the understanding that the seasonal fluctuation of vector-borne diseases in the research area could be traced very clearly. The number of cases was always increasing during the periods of higher temperature and rainfall, and the months right after the

rainy season were generally the ones when the transmission peak occurred. The fact that this pattern entirely depends on weather conditions indicates that climatic factors are the main drivers of disease cycles since they allow more vectors to breed and survive.

The correlation analysis has brought to light that the statistically significant positive relationship between disease incidence and rainfall, as well as average temperature, existed. Relative humidity was also quite strongly associated with the increase in the number of cases. Nevertheless, at very high temperatures, a minor decrease in the number of cases could be detected occasionally, thus suggesting that there may be an upper heat limit at which the survival and reproduction of vectors become less efficient. These findings emphasize the complicated and non-linear aspects of climate–disease interactions in the region.

Moreover, the alterations in time-lagged climatic variables explained a significant proportion of the associations that had been uncovered, which means that changed weather conditions affect disease trends with a time lag of several weeks. Most of the correlations were strongest for the variables that had a lag of one to two months. This period corresponds to the life cycle of mosquito vectors and the incubation period of the pathogens they carry most typically. Hence, it is critical to take delay effects into account when creating predictive models of disease outbreaks.

Besides, the environment and society have remained equally important contributors to the distribution of diseases. Places with high population density, poor drainage, and closeness to water bodies have been the areas where the most cases have been recorded. In the case of urban and peri-urban areas, which are more susceptible than rural ones, the situation is even more dire where wastes are poorly managed, and housing conditions are substandard. Additionally, the areas with more tree cover and surface water have become a higher concentration of diseases which is another indication that the environment is a decisive factor in the determination of vector habitats.

TABLE I DATA DESCRIPTION

Data Type	Source	Time Period	Scale / Variables
Disease data	National Health Suveillance System	[2015–2024]	District/City; Dengue, Malaria, Chikungunya cases
Climate data	Meteorological Department	[2015–2024]	Station-based; Temperature, Rainfall, Humidity
Environmental data	Satelliete imagery (Remote sensing)	[2015–2024]	30–100 m; Vegetation index, water bodies
Population data	Census and urban planning office	Latest census	Ward/District; Density, urbaanization level

Different prediction methods variously achieved success. To a large extent traditional regression models were able to account for the general trends and the seasonal variations, however, they lacked the ability to grasp complex non-linear relationships. On the other side, machine learning algorithms revealed excellent prediction performances and in fact, they were able to reach the highest accuracy level.

IV RESULTS

The tempanalysis of the incidence of diseases has brought to the fore that the seasonal fluctuation of vector-borne diseases in the study area can be very vividly traced. The time of higher temperature and rainfall was always the period when the number of cases was going up, and the months that were right after the rainy season were usually the ones when the transmission peak took place. Since this pattern is entirely dependent on the weather, it can be seen that climatic factors are the main ones that determine cycles of diseases as they make it possible for more vectors to breed and live.The correlation analysis reported that there is a statistically significant positive association between the occurrence of diseases and rainfall, as well as average temperature. Relative humidity also was quite strongly associated with the increase in the number of cases. However, at very high temperatures, occasionally, a slight decrease in the number of cases could be

noticed, thus suggesting that there may be an upper heat limit at which the survival and reproduction of vectors become less efficient. These findings bring to light the complex and non-linear nature of climate–disease interactions in the region.

Moreover, the time changes in lagged climatic variables explained a very large proportion of the associations that had been uncovered, which is quite an indication that the changed weather conditions affect disease trends with a certain time lag. Most of the correlations were strongest for the variables that had a lag of one to two months. This period corresponds to the life cycle of mosquito vectors and the incubation period of the pathogens they carry most typically.

Consequently, it is absolutely essential that delay effects are considered in predictive models of disease outbreaks. Besides that, the environment and society have been equally significant factors that have contributed to the spread of diseases. Places with high population density, bad drainage, and close to water bodies have been areas where the most cases have been recorded.

ith regard to urban and peri-urban areas, which are more vulnerable than rural ones, the situation is worse where wastes are improperly managed, and housing conditions are substandard. Besides that, the areas that have more tree cover and surface water have become areas with a higher concentration of diseases which is yet another indication that the environment is a decisive factor in the determination of vector habitats.

TABLE II YEAR-WISE DISEASE INCIDENCE

Year	Dengue	Malaria	Chikungunya	Total
2019	1,248	734	389	2,371
2020	1,105	690	421	2,216
2021	1,689	903	558	3,150
2022	2,134	1,164	622	3,920
2023	2,586	1,347	774	4,707

CONCLUSION

This study is a clear example of how global changes in the atmosphere massively influence the resulting changes of the increase in parasitic diseases, which have been the central focus of the study area. In addition to that, the study has discovered that climate-related factors such as temperature, rainfall, and humidity are factors that cause questions about the reproduction of vectors for the spread of diseases, their survival, and the transmission of diseases, thus the increase of these diseases is shown by the records. The existence of seasonal trends, as well as that the responses are delayed in the case of climatic changes, is one of the factors which point to the close relationship between environmental factors and disease dynamics.

Nowadays, the understanding of disease occurrence has been made much deeper through the combination of environmental, socioeconomic, and spatial variables. It was found that the places which have rapid urbanization, bad condition of the roads, presence of water in trenches and other places, high population, and uncontrolled vegetation are very likely to be the areas where diseases break out. This means that the transmission of diseases caused by vectors that carry them is not only a natural phenomenon as the climate has patterns, but the way people live, the manner in which infrastructure has been planned, and the conduct of environmental management have all played the roles in the formation of the diseases.

The main factor that significantly brought the revelation of the results more definite was the leading use of statistics and machine learning for modeling. Traditional models can show general trends, but with advanced machine learning techniques, they can uncover complex and non-linear interactions between variables. Very precise models like Random Forest Gradient Boosting are examples that these methods could be the best for the forecast of disease cases in the areas impacted by climate change. This is closely related to the localization of early warning signals in the model outputs; therefore, these outputs can be utilized for the field in real-time monitoring and prevention systems.

The conclusions of this research call for the extreme necessity of the collaboration of integrated cl...imate-sensitive health planni...Disease surveillance system upgrading, urban infrastructur...Restoration, efficient waste, and water management impleme...ntation, and community awareness elevation, are the major...Furthermore, the use of climate data in day-to-day public hea...to be a win in the preparedness and response strategies. In ...In essence, the paper provides arguments that support the veThe primary point of the

research is the construction of a mult...By continuing through the taking anticipatory actions, humanit...It is mostly due to these sophisticated technologies that the v...Such communities will become less exposed to the phenomenon, their resilience will...Simultaneously, it is also quite possible that these communities...

VI.REFERENCES

1. Githeko, A. K., Lindsay, S. W., Confalonieri, U. E., & Patz, J. A. (2000). Climate change and vector-borne diseases: a regional analysis. Nature, 408, 613–620.

2. Chabot-Couture, G., Nigmatulina, K., & Eckhoff, P. (2014). The environment data that supports vector-borne diseases modeling and epidemiology. PLoS One, 9(4), e94741.

3. Abbasi, E. (2025). Climate change impact on travel-related vector-borne diseases: A case study of dengue virus transmission. Science of The Total Environment, 102841.

4. Teeluck, M., Adegboye, O., Karl, S., Iyaloo, D. P., & McBryde, E. (2025). A Systematic Review to Comprehend the Impact of Climate Change on the Re-Emergence of Mosquito-Borne Diseases in Small Island Nations. International Journal of Environmental Research and Public Health, 18(3), 223–247.

5. Holly Jarman and Phillip M. Singer, Jarman, H., Singer, P. M., Holmes, I. A., Hsu, J., Harper, C., Unwala, N., & Willison, C. E. (2025). Draining the Swamp: The Local Governance of Mosquito Borne Diseases in Florida. Ecology and Society, 30(1), 30.

6. Andre B. B. Wilke, Priscilla Farina, and Marco Ajelli,´ Wilke, A. B., Farina, P., Ajelli, M., Canale, A., Dantas-Torres, F., Otranto, D., & Benelli, G. (2025). Human migrations, anthropogenic changes, and insect-borne diseases in Latin America. Scientific Reports, 13, 1–4.

7. Eduardo C. Araujo, Claudia T. Codec¸o and Sandro´ Loch, Araujo, E. C., Codec¸o, C. T., Loch, S., Vacaro, L. B., Freitas, L. P., Lana, R. M., ... & Coelho, F. C. (2025). Large-scale epidemiological modelling: scanning for mosquito-borne diseases spatio-temporal patterns in Brazil. Computers in Biology and Medicine, 135, 241261.

8. Jonas E. Mertens, Mertens, J. E. (2024). A history of malaria and conflict. Medical History, 68(3), 165.

9. Muhammad Ammar, Ammar, M., Moaaz, M., Yue, C., Fang, Y., Zhang, Y., Shen, S., & Deng, F. (2025). Emerging arboviral diseases in Pakistan: Epidemiology and public health implications. Science of The Total Environment, 908, 232.

10. Luigi Cofone, Marise Sabato and Carolina Di Paolo, Cofone, L., Sabato, M., Di Paolo, C., Di Giovanni, S., Donato, M. A., & Paglione, L. (2025). Urban, architectural, and socioeconomic factors contributing to the concentration of potential arbovirus vectors and arbovirosis in urban.

11. Yavari Nejad, F., & Varathan, K. D. (2021). Identification of significant climatic risk factors and machine learning models in dengue outbreak prediction. International Journal of Environmental Science and Technology, 18(1), 141.

12. Leung XY, et al. A comprehensive review on dengue outbreak prediction models: existing situation and upcoming perspectives. BMJ Open. 2023;13(2):e0010631.

13. Kumar, S., Srivastava, A., & Maity, R. (2024). Modeling climate change impacts on vector-borne disease using machine learning models: Case study of Visceral leishmaniasis (Kala-azar) from Indian state of Bihar. Computers in Biology and Medicine, 154, 121490.

14. Patil, S., & Pandya, S. (2021). Forecasting dengue hotspots associated with variation in meteorological parameters using regression and time series models. International Journal of Advanced Research, 9(3), 798034.

15. Zheng, J. X., Lu, S. N., Li, Q., Li, Y. J., Xue, J. B., Gavana, T., ... & Zhou, X. N. (2025). Deciphering the climate-malaria nexus: A machine learning approach in rural southeastern Tanzania. PLoS One, 14, 124–130.

16. Hussain-Alkhateeb, L., Rivera Ramirez, T., Kroeger, A., Gozzer, E., & Runge-Ranzinger, S. (2021). Early warning systems (EWSs) for chikungunya, dengue, malaria, yellow fever, and Zika outbreaks: What is the evidence? A scoping review. PLOS Neglected Tropical Diseases, 15, (9), e0009686.

17. Leon, M. I., Iqbal, M. I., Meem, S., Alahi, F., Ahmed, M., Shatabda, S., & Mukta, M. S. H. (2021, December). Dengue outbreak prediction from weather aware data. In IISA (pp. 1–11). Cham: Springer International Publishing.

18. Chowdhury, A. H., & Rahman, M. S. (2025). Machine learning and spatio-temporal analysis of meteorological factors on waterborne diseases in Bangladesh. International